CS 461: Machine Learning
Assignment #1: Exploratory data analysis (EDA)

This assignment will focus explicitly on describing a dataset, and data visualization, and include more of an emphasis on **numpy, pandas, matplotlib and seaborn** libraries.

You can complete the assignment using either Jupyter Notebooks or VS code.

Part-1: Machine Learning Libraries

There is a rich number of tools and open source libraries available to facilitate practical machine learning..

1. Analyze the trade-offs between Jupyter Notebooks, Visual Studio Code, and Google Colab. Which tool is best suited for an independent data scientist versus a collaborative machine learning team? In addition, create a comparison table highlighting how the tools differ in compute infrastructure, Git version control, and production scalability for machine learning?
2. What is a machine learning library? Explain in details some libraries such as **numpy, pandas, matplotlib and seaborn** libraries.
3. Explain the concept of vectorization in NumPy. Why is vectorization preferred over standard Python for loops when handling large machine learning datasets? Provide a short code comparison showcasing the execution time difference.
4. Explain the concept of a DataFrame in Pandas and provide a Python code example of its implementation. Additionally, discuss the key methods attached to this class that a data analyst would use for exploration, cleaning missing data, and transforming datasets. (You can create a table for your answer)

Part-2: Exploratory data analysis (EDA)

Cell phones are now essential everyday tools, and their prices vary widely depending on their features. This assignment tasks you with investigating the variables that influence mobile phone pricing and using data analysis to explore the specific features that define different phone samples.

1- Understanding the data

1. Download the dataset from MS-Teams and upload to Jupyter Notebook or your Drive.
2. Import the necessary libraries for this assignment, then write a report about the dataset, such as:
   a. How many entries does it have?
   b. How many features does it have?
   c. Does it contain any categorical data?
   d. Generate and interpret the descriptive statistics of the dataset (e.g., mean, median, standard deviation, min, max) using the describe() method
   e. Check and remove the outliers in the dataset. Explain the process and your data after and before removing the outliers.
   f. Check and handle any missing values in the dataset. Explain the process and your data.
   g. Create a new engineered feature named screen_area by multiplying the screen width (sc_w) and screen height (sc_h). Check if this newly created feature has a stronger correlation with price_range than the individual width and height features.
   h. Apply **Min-Max Scaling** to scale all numerical features strictly between the ranges of 0 and 1. Explain your process. And show the data before and after the process.
   i. Explain why feature normalization (step j) should ideally be fitted only on a training dataset rather than the entire dataset during a real-world machine learning pipeline\steps. What risk do we run if we normalize everything together first?
   j. Create a count plot of the 'price_range' column and label the bars with their counts
   k. Analyze the distribution of binary features (like dual_sim, blue, touch_screen, and wifi) across different price_range tiers. For example, calculate whether phones with or without Bluetooth have a higher frequency in each price category
   l. Use the corr() method and report the columns that are closely related to each other. Provide some evidences (e.g., bar chart, statics) that support your claims.

m. Is the target variable (price_range) balanced or imbalanced? Explain how an imbalanced target variable might negatively impact the training phase of a machine learning classifier.

n. Identify two features in the dataset that exhibit a strong negative correlation. Based on real-world smartphone manufacturing and consumer habits, explain the practical reason behind this negative relationship.

AI Assistance: (For more information check the syllabus)

- Submitting any generative artificial intelligence (AI), such as ChatGPT content, as your own work/words (e.g., cutting and pasting content into written assignments) will result - at a minimum - in a **ZERO** on the assignment.
- Include detailed inline comments explaining your code.

Submission

In order to submit your assignment, please upload the following files to MS Teams:

1. Assignment-1_Lastname_FirstName.pdf